Senior Project Proposal

Janiyah M. Cutchin-Brown

Charleston Southern University

Bachelor of Science in Computer Science

Professor Michael O'Neill

May 2027

Problem Statement

Many individuals aim to manage their health through improved nutrition, whether for weight management, athletic performance, or general wellness. In an ideal state, users would be able to easily track daily calorie intake, understand how food choices align with personal goals, and observe measurable progress over time through a clear and intuitive system. Tracking would be simple, efficient, and sustainable as part of a daily routine.

However, existing calorie tracking applications are often cluttered with excessive features, complicated interfaces, or subscription-based restrictions that reduce accessibility. Research in behavioral health and habit formation suggests that overly complex systems increase cognitive load and reduce long-term adherence to tracking behaviors. When users find applications overwhelming or time-consuming, they are more likely to discontinue use, ultimately undermining their health goals. Given that consistent self-monitoring is strongly associated with improved weight management outcomes, the lack of a streamlined and user-centered tracking solution presents a meaningful gap.

This project proposes the design and development of a web-based calorie tracking application that prioritizes simplicity, usability, and workflow clarity. The solution will allow users to record meals, view daily calorie totals, and monitor basic trends through a clean interface without unnecessary complexity. By reducing friction in daily nutrition logging, the proposed system aims to promote consistency and long-term engagement. In summary, this project addresses the usability limitations of current calorie tracking tools by designing a simplified and accessible system that supports sustainable health monitoring.

Project Description

This project involves the design and implementation of a web-based calorie tracking application. The system will allow users to:

- Create and manage personal accounts
- Log daily food intake with associated calorie values
- View total daily calorie consumption
- Monitor calorie trends over time
- Set basic calorie goals
- Visualize progress through graphical representations

The application will follow a client-server architecture with a frontend interface communicating with a backend API connected to a relational database. Emphasis will be placed on usability, structured data storage, and clear visual feedback.

The final deliverable will include a functional prototype, documented system architecture, database schema, API documentation, and a formal presentation.

Proposed Implementation Language(s)

- Frontend: JavaScript (React.js)
- Backend C# (.NET Web API)
- Database SQL (SQL Server)

Libraries, Packages, Development Kits

Frontend

- React.js
- Chart.js
- Axios

Backend

- ASP.NET
- Entity Framework Core
- JWT Authentication

Database

- Microsoft SQL Server

Developmental Tools

- Visual Studio
- Visual Studio Code
- Github

Additional Software/Equipment Needed

- Development computer capable of running VSC and SQL Server
- Local web server environment
- Github repository
- Internet access

No specialized hardware is required

Alternative Solutions and Rationale

Alternative solutions include:

1. Developing a mobile application instead of a web application.

2. Integrating third-party nutrition databases (e.g., API-based food lookup).

3. Using a full-stack JavaScript solution (Node.js backend instead of .NET).

The chosen approach (React frontend with .NET backend) was selected because it aligns with coursework experience and strengthens backend API development skills. A web-based application ensures accessibility across devices without requiring mobile platform-specific development. Additionally, limiting integration with external food APIs maintains project scope and emphasizes core system design rather than data sourcing complexity.

Personal Motivation

This project aligns with both academic and personal interests in health technology and software usability. Developing a calorie tracking system will strengthen knowledge in full-stack development, database design, API construction, and user-centered interface design.

Additionally, this project provides practical experience in designing systems that address real-world behavioral challenges, particularly in habit formation and user engagement. By completing this project, the student will enhance skills in structured system architecture, data modeling, and scalable web application development, preparing for professional software engineering roles.

Outline of Future Research Efforts

To complete this project, the following steps will be taken:

1. Conduct research on user interface best practices for habit-forming applications.

2. Analyze existing calorie tracking systems to identify usability gaps.

3. Design system architecture diagrams and database schemas.

4. Implement backend API endpoints for user management and calorie logging.

5. Develop frontend components and integrate them with backend services.

6. Perform system testing and debugging.

7. Conduct usability evaluation and refinement.

8. Prepare final documentation and presentation materials.

Expected deliverables include:

- Functional prototype
- System architecture diagram
- Database schema diagram
- API documentation
- Final written report
- Presentation/demo

CSCI 498 (Senior Project I)

Date	Task
Week 1–2	Finalize proposal and requirements
Week 3–4	System architecture and database design
Week 5–6	Backend API development (core endpoints)
Week 7–8	Initial frontend implementation
Week 9	Integration testing
Week 10	Mid-project review with advisor

CSCI 499 (Senior Project II)

Date	Task
Week 1–3	Feature completion and refinement
Week 4–6	Graph visualization implementation
Week 7	System testing and debugging
Week 8	Documentation completion
Week 9	Final presentation preparation
Week 10	Project demonstration and submission

FR-01

ID: FR-01

Type: Functional

Description:

The system shall allow a user to create a personal account by providing a unique email address and password.

Rationale:

Users must have individual accounts to securely store and retrieve their personal calorie data.

Fit Criterion:

A user can successfully create an account when providing:

A unique email address

A password meeting defined security rules

The system prevents registration if:

The email already exists

Required fields are blank

Account creation completes within 3 seconds under normal system load.

Priority: Must

Dependencies: None

FR-02

ID: FR-02

Type: Functional

Description:

The system shall allow a registered user to log in using their email address and password.

Rationale:

Authenticated access is required to protect personal health data and provide access to stored information.

Fit Criterion:

A user with valid credentials is granted access to their dashboard.

A user with invalid credentials receives an error message.

Login response time shall not exceed 2 seconds under normal operating conditions.

After 5 consecutive failed login attempts, the account shall be temporarily locked for 5 minutes.

Priority: Must

Dependencies: FR-01

FR-03

ID: FR-03

Type: Functional

Description:

The system shall allow a logged-in user to record a food entry including food name, calorie amount, and date of consumption.

Rationale:

The primary purpose of the system is to enable users to track daily calorie intake.

Fit Criterion:

A user can submit a food entry containing:

Food name (required)

Calorie value (required, numeric, greater than 0)

Date (defaults to current date if not specified)

The system stores the entry in the database.

The entry appears in the user's daily log immediately after submission.

Invalid calorie values (negative or non-numeric) are rejected.

Priority: Must

Dependencies: FR-02

FR-04

ID: FR-04

Type: Functional

Description:

The system shall calculate and display the total daily calorie intake for the selected date.

Rationale:

Users need immediate feedback on daily consumption to compare against their calorie goals.

Fit Criterion:

The total displayed equals the sum of all calorie entries for that date.

The calculation updates automatically within 1 second after a new entry is added or removed.

The total resets correctly when a different date is selected.

Priority: Must

Dependencies: FR-03

FR-05

ID: FR-05

Type: Functional

Description:

The system shall allow users to set and update a personal daily calorie goal.

Rationale:

Goal-setting supports behavior change and allows users to measure progress.

Fit Criterion:

A user can enter a numeric daily calorie goal greater than 0.

The goal is saved to the user's profile.

The dashboard displays:

Current daily total

Goal value

Remaining calories (goal − total)

Changes to the goal are reflected immediately after saving.

Priority: Must

Dependencies: FR-02

FR-06

ID: FR-06

Type: Functional

Description:

The system shall allow a user to edit a previously recorded food entry.

Rationale:

Users may enter incorrect calorie values or food names and need the ability to correct entries.

Fit Criterion:

A user can modify the food name, calorie value, or date of an existing entry.

Changes are saved to the database immediately after confirmation.

The updated value is reflected in the daily calorie total within 1 second.

Priority: Must

Dependencies: FR-03, FR-04

FR-07

ID: FR-07

Type: Functional

Description:

The system shall allow a user to delete a previously recorded food entry.

Rationale:

Users need the ability to remove incorrect or unwanted entries.

Fit Criterion:

A user can select and delete an entry from their daily log.

The system removes the entry from the database.

The daily calorie total updates within 1 second after deletion.

Priority: Must

Dependencies: FR-03, FR-04

FR-08

ID: FR-08

Type: Functional

Description:

The system shall allow users to view a history of their calorie intake across multiple days.

Rationale:

Users need to review past behavior to evaluate consistency and progress toward health goals.

Fit Criterion:

Users can select a date within the stored history.

The system displays all entries and the total calories for that date.

Historical data retrieval occurs within 2 seconds.

Priority: Must

Dependencies: FR-03

FR-09

ID: FR-09

Type: Functional

Description:

The system shall display graphical visualizations of calorie trends over time.

Rationale:

Visual feedback helps users quickly understand patterns in their calorie intake.

Fit Criterion:

The system generates a graph displaying daily calorie totals over a selected time range.

Users can view at least 7, 14, or 30 days of data.

Graphs render within 2 seconds after selection.

Priority: Should

Dependencies: FR-08

FR-10

ID: FR-10

Type: Functional

Description:

The system shall allow users to log out of their account.

Rationale:

Logging out ensures user sessions can be securely terminated, particularly on shared devices.

Fit Criterion:

Selecting “Logout” immediately terminates the user session.

The system redirects the user to the login page within 2 seconds.

Attempting to access protected pages after logout requires re-authentication.

Priority: Must

Dependencies: FR-02

US-01

ID: US-01

Type: Usability – Ease of Use

Description:

The system interface shall allow users to record a food entry in three steps or fewer.

Rationale:

Reducing interaction complexity encourages consistent daily tracking.

Fit Criterion:

A user can log a food entry using no more than three user actions (e.g., open entry form, input data, submit).

In usability testing, at least 80% of users successfully complete the task without assistance.

Priority: Must

Dependencies: FR-03

US-02

ID: US-02

Type: Usability – Accessibility

Description:

The system shall support accessibility guidelines to ensure readability and navigation for diverse users.

Rationale:

Accessible design improves usability for individuals with visual or cognitive impairments.

Fit Criterion:

Text contrast ratio meets WCAG 2.1 Level AA standards.

All primary features are accessible using keyboard navigation.

Screen reader software can correctly interpret navigation elements.

Priority: Should

Dependencies: None

LF-01

ID: LF-01

Type: Look and Feel – Appearance

Description:

The system interface shall present calorie totals, goals, and logs in a visually organized dashboard layout.

Rationale:

Clear visual organization helps users quickly interpret their daily progress.

Fit Criterion:

The dashboard displays:

Daily calorie total

Calorie goal

Remaining calories

These elements are visible without scrolling on standard desktop resolutions (1920×1080).

Priority: Should

Dependencies: FR-04, FR-05

PE-01

ID: PE-01

Type: Performance – Speed and Latency

Description:

The system shall respond to user actions within acceptable performance limits.

Rationale:

Slow systems reduce user satisfaction and discourage continued use.

Fit Criterion:

95% of system requests complete within 2 seconds under normal load conditions.

Page load time does not exceed 3 seconds.

Priority: Must

Dependencies: None

PE-02

ID: PE-02

Type: Performance – Capacity

Description:

The system shall support multiple concurrent users without performance degradation.

Rationale:

The system must remain usable if multiple users access it simultaneously.

Fit Criterion:

The system supports at least 100 concurrent active users.

Average response time remains under 3 seconds during load testing.

Priority: Should

Dependencies: PE-01

SEC-01

ID: SEC-01

Type: Security – Access

Description:

The system shall restrict access to personal calorie data to authenticated users only.

Rationale:

User health data must remain private and protected from unauthorized access.

Fit Criterion:

Users cannot access calorie data without logging in.

Attempting to access protected endpoints without authentication returns an HTTP 401 or equivalent error.

Priority: Must

Dependencies: FR-02

SEC-02

ID: SEC-02

Type: Security – Privacy

Description:

The system shall protect user credentials during authentication.

Rationale:

Protecting login credentials prevents unauthorized account access.

Fit Criterion:

Passwords are never stored in plain text.

Authentication requests are transmitted using HTTPS.

Security testing confirms credentials cannot be retrieved from the database in readable form.

Priority: Must

Dependencies: FR-01, FR-02

REL-01

ID: REL-01

Type: Reliability and Availability

Description:

The system shall maintain continuous availability during normal operating hours.

Rationale:

Users rely on consistent access to log daily meals.

Fit Criterion:

System uptime shall be at least 99% during operational periods.

In the event of failure, the system restores service within 10 minutes.

Priority: Should

Dependencies: None

REL-02

ID: REL-02

Type: Robustness / Fault Tolerance

Description:

The system shall handle invalid input without crashing or corrupting stored data.

Rationale:

Robust error handling prevents data loss and ensures system stability.

Fit Criterion:

Invalid input (e.g., text in numeric calorie fields) is rejected with a clear error message.

The system continues functioning normally after invalid input is submitted.

No database corruption occurs during testing.

Priority: Must

Dependencies: FR-03

AUD-01

ID: AUD-01

Type: Security – Audit

Description:

The system shall record user actions in an audit log to support security monitoring and system troubleshooting.

Rationale:

Maintaining audit logs enables detection of unauthorized activity, supports debugging, and provides traceability of user actions.

Fit Criterion:

The system records the following events:

Successful and failed login attempts

Creation of food entries

Modification of food entries

Deletion of food entries

Each audit record includes:

User identifier

Action performed

Timestamp of the action

Audit logs are stored for a minimum of 30 days.

Audit records can be retrieved within 2 seconds during testing.

Priority: Should

Dependencies: FR-02, FR-03, FR-06, FR-07